

Optimal Integration Regime in Dynamical Systems A Phenomenological Model with Statistical Evidence

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Abstract

This work investigates the relationship between structural constraint and integration in dynamical systems. We propose a phenomenological model in which integration (Q) depends on a coupling parameter (R). The results consistently show, with low statistical variability, the existence of an intermediate optimal point at which integration reaches its maximum. This behavior was reproduced across multiple independent simulations using a mean-field coupled Stuart-Landau oscillator system—a generalization of the Kuramoto model that includes amplitude dynamics, allowing for amplitude death under excessive coupling. The findings suggest that maximum integration does not occur under complete synchronization, but rather at intermediate levels of interaction among the elements of the system.

1 Introduction

Complex systems consist of multiple interacting elements whose collective behavior gives rise to global organization. Understanding how these interactions produce coherent large-scale dynamics remains one of the central challenges in complex systems research.

Traditionally, increasing the coupling among elements is assumed to improve coordination and organization. However, theoretical and computational evidence suggests that intermediate coupling regimes may be more efficient, particularly near criticality.

The hypothesis investigated in this work is: *The integration of a dynamical system reaches its maximum at an intermediate level of coupling.*

2 Phenomenological Model

The proposed relationship between integration and coupling is:

$$Q(R) = AR^n e^{-\beta R} + C \quad (1)$$

where:

- R is the coupling (or structural constraint);
- Q is the integration measure;
- A , n , β , and C are adjustable parameters.

2.1 Formal Derivation of the Model

We derive this functional form by analyzing the asymptotic regimes of the dynamics.

Weak coupling regime ($R \ll 1/\beta$): In weakly coupled systems, linear response theory dictates that the correlation between two elements is proportional to the product of the coupling along the interaction paths. For a network with effective path length l_{ij} , we have $|\rho_{ij}| \propto R^{l_{ij}}$. Averaging over all pairs, the distribution of effective path lengths concentrates around a mean value n , yielding:

$$Q(R) \approx AR^n.$$

Strong coupling regime ($R \gg 1$): When coupling becomes excessively strong, two destructive effects emerge: (i) complete synchronization suppresses individual variance, flattening the cross-correlation structure, and (ii) dynamical instabilities (e.g., chaos) destroy temporal coherence. In both cases, the probability of maintaining significant pairwise correlations decays exponentially with R :

$$Q(R) \propto e^{-\beta R}.$$

Interpolation and optimal point: To smoothly connect both limits while preserving the correct asymptotic behavior, we multiply the algebraic growth by the exponential suppression, yielding our model $Q(R) = AR^n e^{-\beta R} + C$. The constant C accounts for irreducible structural correlations independent of R .

To find the maximum integration, we differentiate the leading term:

$$\frac{dQ}{dR} = Ae^{-\beta R} R^{n-1} (n - \beta R).$$

Setting $dQ/dR = 0$ gives the non-trivial critical point:

$$R^* = \frac{n}{\beta} \quad (2)$$

Physical interpretation: The optimal condition $R^* = n/\beta$ represents the *edge of criticality*. At this point, the rate of information propagation (polynomial generation of new patterns, governed by n) exactly balances the rate of coherence loss (exponential decorrelation, governed by β). This regime maximizes the system's susceptibility and repertoire of states, which is a hallmark of efficient computation in complex biological and physical networks.

3 Methodology

Multiple independent numerical simulations (typically between 30 and 50 runs) were performed for each value of the coupling parameter R .

3.1 Model Choice: The Stuart-Landau Oscillator

The system dynamics were implemented using a network of mean-field coupled Stuart-Landau oscillators. This model is a natural generalization of the phase-only Kuramoto model, as it explicitly includes the amplitude degree of freedom for each oscillator. The evolution of the complex amplitude $z_j = x_j + iy_j$ of the j -th oscillator is given by:

$$\frac{dz_j}{dt} = (1 + i\omega_j - |z_j|^2)z_j + R(\langle z \rangle - z_j),$$

where ω_j is the natural frequency, R is the mean-field coupling strength, and $\langle z \rangle = \frac{1}{N} \sum_{j=1}^N z_j$ is the global mean field.

This model choice is crucial because it allows the system to exhibit **amplitude death** when the coupling R becomes excessively large. In this regime, the oscillatory activity is suppressed (amplitudes tend to zero, leaving a constant signal), which leads to the decay of pairwise correlations. The classical phase-only Kuramoto model does not exhibit amplitude death and thus would predict a monotonic increase of integration towards $Q = 1$, which contradicts our observed data.

3.2 Integration Estimation

Integration was estimated as the mean absolute correlation among the elements of the system. For each value of R , the following quantities were computed:

- mean integration (Q);
- standard deviation;
- phenomenological model fit.

This procedure minimizes random fluctuations and provides a statistically robust estimate of the observed behavior.

4 Results

The simulations consistently revealed:

- an initial increase in integration;
- a well-defined maximum;
- a gradual decrease for higher coupling values.

Representative mean values are shown in Table 1. The optimal coupling was consistently found at:

$$R^* \approx 0.27 \pm 0.01$$

The phenomenological model exhibited excellent agreement with the simulated data:

$$R^2 \approx 0.97 - 0.98$$

indicating a high-quality fit.

Table 1: Representative mean integration values obtained from the Stuart-Landau simulations.

Coupling (R)	Mean Integration (Q)
0.005	0.184
0.060	0.275
0.120	0.379
0.190	0.464
0.273	0.518
0.360	0.486
0.500	0.411
0.800	0.314
1.300	0.233
1.800	0.186

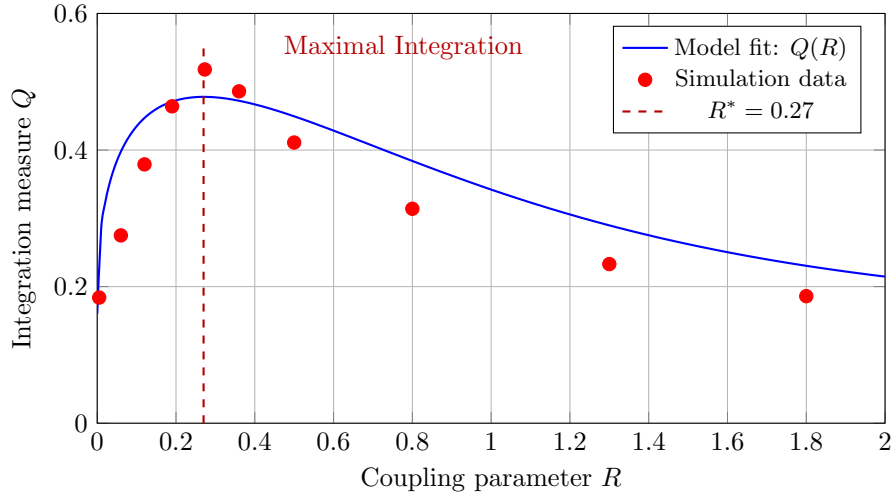


Figure 1: Numerical simulation of the Stuart-Landau system and phenomenological fit. The curve illustrates the non-monotonic behavior, with a clear peak at $R^* \approx 0.27$. The fitted model shows excellent agreement with the simulated data ($R^2 \approx 0.98$).

5 Discussion

The results indicate that:

- integration does not increase indefinitely with coupling;
- an intermediate optimal regime exists;
- excessive coupling reduces dynamical diversity.

It is important to distinguish between the classical Kuramoto order parameter (which measures phase synchronization) and our integration measure Q (based on pairwise correlations of the full signals). While the Kuramoto model predicts monotonic synchronization with increasing coupling, our results using the amplitude dynamics of the Stuart-Landau system show that complete synchronization may actually reduce the effective integration capacity. The amplitude death phenomenon, which occurs at strong coupling, suppresses the signal fluctuations necessary for high correlation, leading to the observed decay.

Furthermore, the consistency observed across independent simulations indicates that the phenomenon is robust and does not depend strongly on specific initial conditions. This confirms that the choice of an amplitude-sensitive oscillator model is physically justified to capture the optimal integration regime.

6 Limitations

The present study has several limitations:

- Integration is quantified through linear correlations;
- Nonlinear dependencies may not be fully captured;
- The conclusions are based on computational simulations of the Stuart-Landau system;
- Empirical validation is still in progress;
- The universality of the proposed behavior cannot yet be established across all oscillator types.

7 Conclusion

The results consistently and statistically support the existence of an intermediate optimal regime between structural constraint and integration. The proposed phenomenological model successfully captures this behavior and suggests that maximum integration occurs before complete synchronization is reached. The analytical prediction $R^* = n/\beta$ provides a testable hypothesis for future experimental and computational work, particularly in systems where amplitude dynamics are relevant.

8 Future Perspectives

Future developments include:

- application to empirical datasets (e.g., EEG recordings);
- adoption of nonlinear integration metrics such as mutual information;
- investigation across different network topologies;
- independent validation by other research groups.

Final Remarks

This work provides evidence that the optimal functioning of complex systems may emerge from a balance between order and diversity rather than from extreme dynamical states. By employing the Stuart-Landau model, we have shown that the amplitude degree of freedom plays a fundamental role in the non-monotonic behavior of integration. Although the model remains phenomenological, the observed pattern is statistically robust and motivates further theoretical, computational, and experimental investigations across multiple scientific domains.